NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC4000 - Machine Learning

American Express - Default Prediction Project

Group 37

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TABLE OF CONTENTS

1. Introduction and Project Aim
    1.1 Project Outline and Aim
    1.2 Competition Description
2. Exploratory Dataset Analysis
    2.1 Positive Clients
    2.2 S_2 Statement Count
    2.3 Categorical Features
    2.4 Numerical Data
    2.5 Histogram Data
    2.6 Next Steps
3. Modelling Approach
    3.1 Models Proposed
    3.2 Models Used
4. Data Preprocessing
    4.1 Analysis
    4.2 Our Unique Solution
    4.2 Data Splitting
5. Model Training and Evaluation
    5.1 Model 1 (Random Forest)
    5.2 Model 2 (LightGBM)
6. Results & Comparison
    6.1 Dataset Characteristic
    6.2 Linear vs Non-Linear
    6.3 LightGBM Results
    6.4 Random Forest Results
7. Conclusion
8. Reflection
9. Future Improvements
References

1. Introduction and Project Aim

1.1 Project Outline and Aim

This project tackles the *American Express – Default Prediction* Kaggle competition, where the goal is to predict whether a customer will default on their credit obligations in the future based on a large set of historical statements. Our aim is to design a solution that is both *accurate and computationally efficient*: we build a memory-safe "last-statement" feature pipeline and train machine learning models to compare their performance under the official AMEX evaluation metric.

1.2 Competition Description

Debit and credit cards have become ubiquitous to our everyday life, overtaking cash as a common payment method due to its convenience and accessibility. As such, it has become a priority for companies like American Express to know that their credit card users will be able to adequately pay back their loan in full. A machine learning model that can answer this question could end up deciding whether to approve or deny new requests for credit cards, based solely on a customer's spending profile.

This project tasks us with successfully predicting whether a customer will default on their credit card payments, given a wide variety of anonymized features. We will make use of different machine learning models to observe how we can accurately provide a prediction on the factors that affect and shape the tendencies for customers to pay back what they owe to American Express on time.

2. Exploratory Dataset Analysis

Before any machine learning is attempted, we must understand the data. Upon simple observation, we can observe the following key insights for the dataset:

Metrics	**Data**
Number of columns detected	190
Number of column in *float64*	185
Number of column in *object*	4
Number of column in *int64*	1
Number of row entries	5,531,451
Number of distinct *customerID*	458,913

Of which, below are the row entries that contain the majority "NA" as shown below in the table from the sampling row of 1,000,000.

	NA% (approx from sample)		**NA% (approx from sample)**
D_87	99.93	**R_9**	94.31
D_88	99.88	**B_29**	93.05
D_108	99.47	**D_106**	90.22
D_110	99.41	**D_132**	90.19
D_111	99.41	**D_49**	90.14
B_39	99.37	**R_26**	89.02
D_73	99.02	**D_76**	88.74
B_42	98.70	**D_66**	88.67
D_137	96.47	**D_42**	85.70
D_134	96.47	**D_142**	82.88
D_136	96.47	**D_53**	73.88
D_135	96.47	**D_82**	73.36

D_138	96.47		

It can be noted that the variables in this dataset are anonymized: that is, what they specifically represent is left unknown. We are only given general categories like “delinquency” and “risk”, with an understanding that features with the same category should behave similarly as predictors of customer defaulting.

We will now perform some exploratory analysis on the dataset to better understand the data distribution.

2.1 Positive Clients

Train customers: 458913
Default rate (target=1): 0.2589

target	count
0	340085
1	118828

We can observe that 75% of the subjects make their payments on time, with 25% failing to pay American Express back.

```
labels["target"].value_counts().plot(kind="bar", color=["steelblue","salmon"])
plt.title("Target Distribution (0 = No Default, 1 = Default)")
plt.xlabel("Target")
plt.ylabel("Count")
plt.show()
```

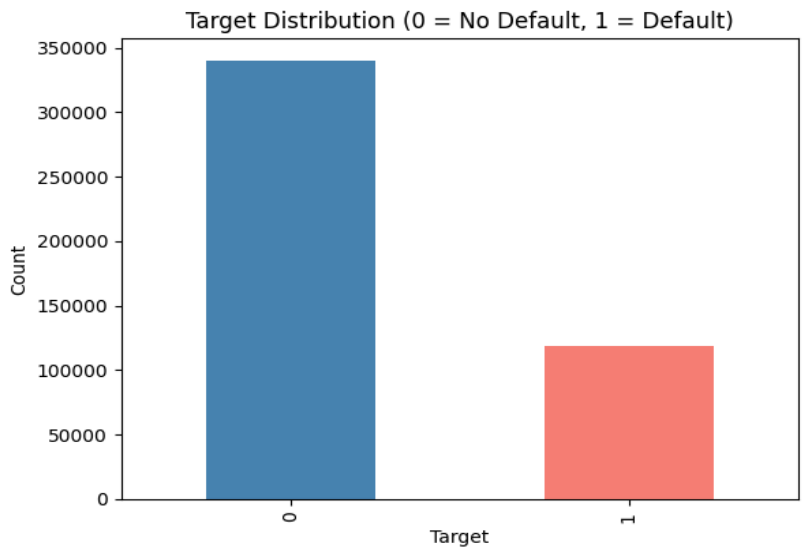
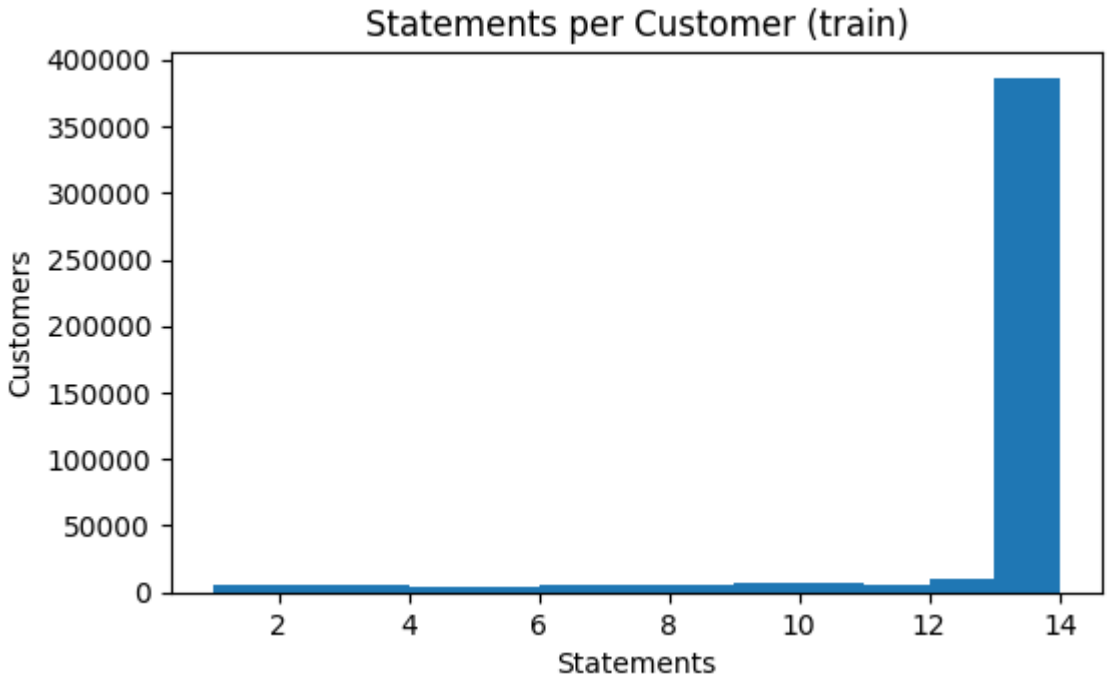


Fig: Target Imbalance bar Chart

2.2 S_2 Statement Count

It can be observed from the data that out of 458,913 *customer_ID*, they collectively have an average of 12.05 statements with a median count of 13 statements.



2.3 Categorical Features

This report also explores some categorical features of the dataset given.

Top values for D_63:
CO 744637
CR 168470
CL 78784
XZ 4918
XM 1919
XL 1272
Name: count, dtype: int64

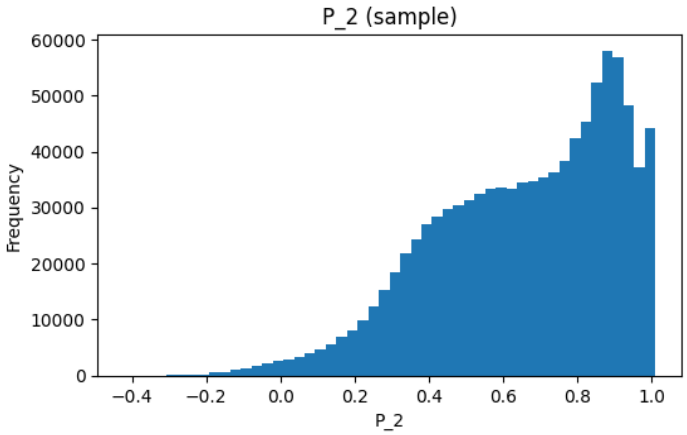
Top values for D_64:
O 527342
U 275718
R 151483
NaN 39038
-1 6419

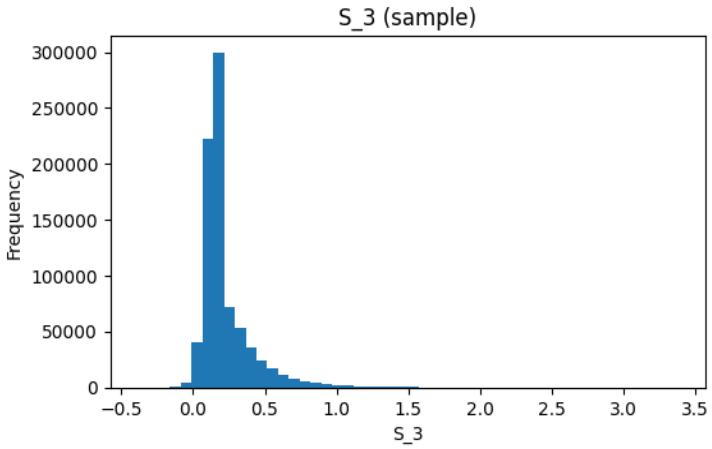
Name: count, dtype: int64

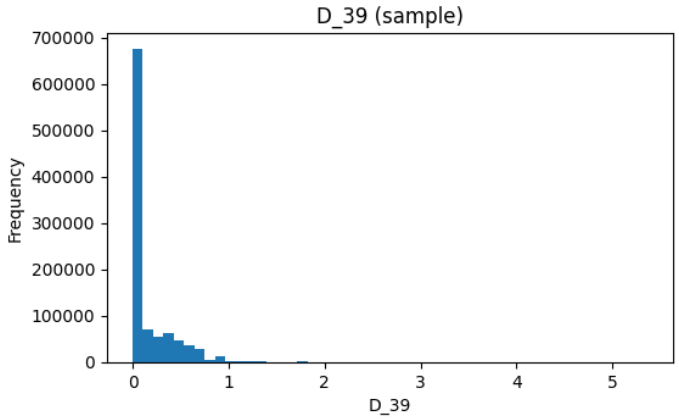
From the data shown above, we can see that D_63 is highly likely to be the types of transactions that are made using the American Express card. We can see that "CO" for "Charge-Off" and "CR" for "Credit" are the most often used categories for column ***D_63*** while "O" and "U" are the 2 highest used categories for column ***D_64***.

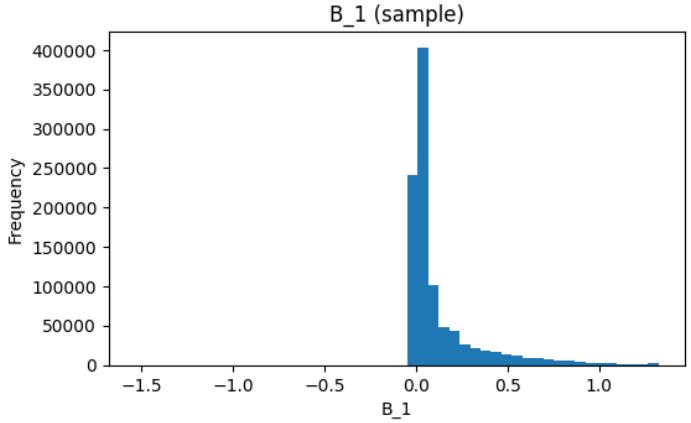
2.4 Numerical Data

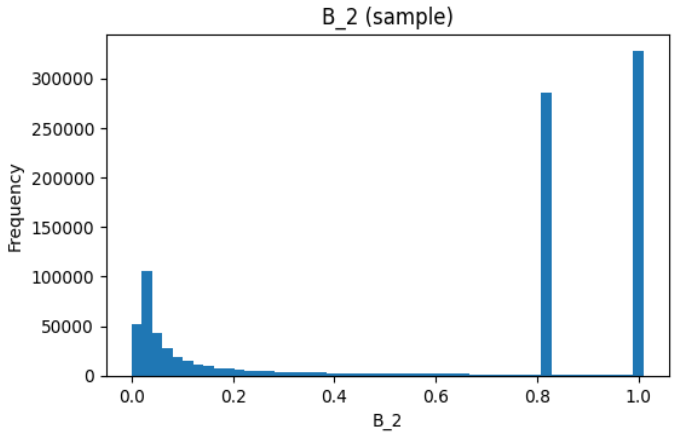
It can also be observed that there are not just categorical data columns, but numerical float data columns too. Below are some bar charts to show the distribution of the numerical data columns.

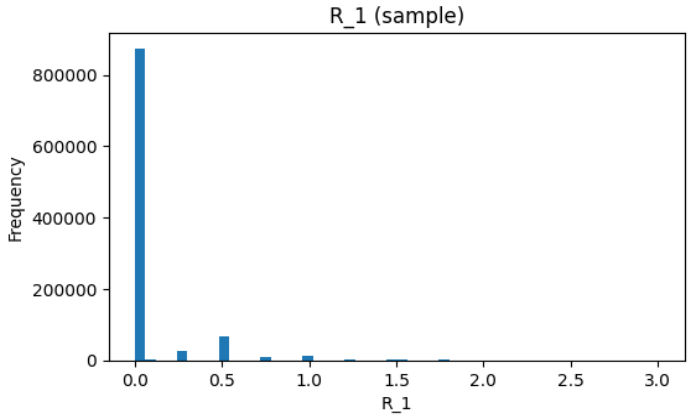




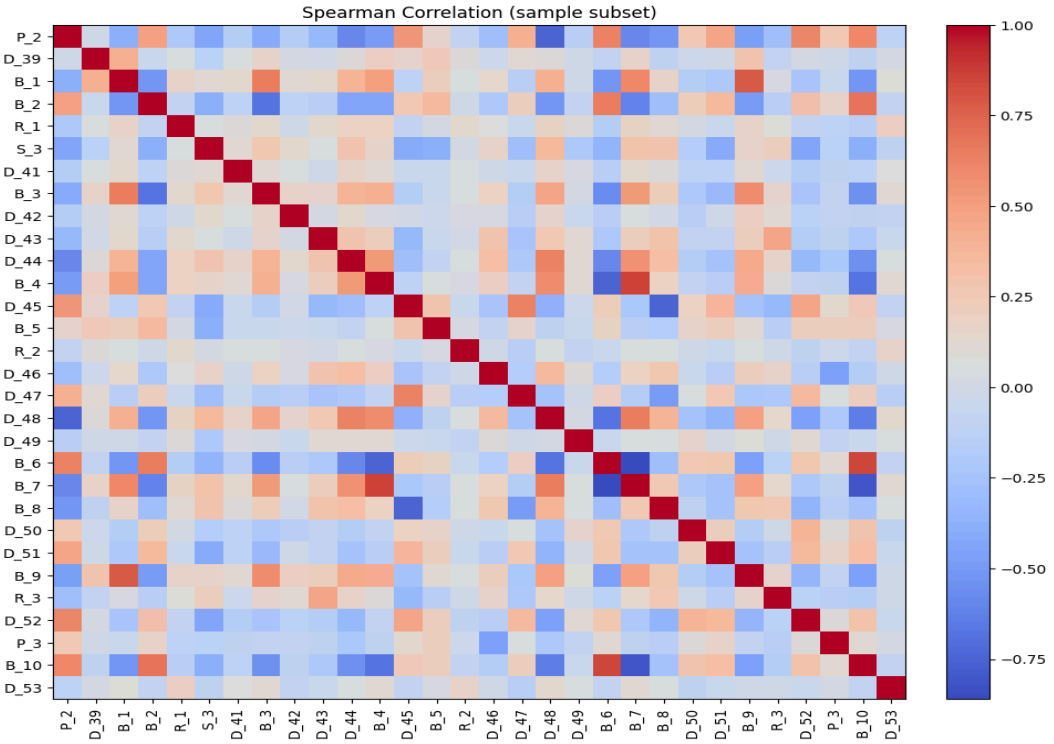








Next, we also use a histogram to determine if there is any correlation between any of the data.



On further inspection, we can observe and list down the 2 highest correlation dataset.
They are ***B_4*** and ***B_7*** with a correlation value of **0.864** as well as ***B_6*** and ***B_10*** with a correlation value of **0.847**.

2.6 Next Steps

With some analysis of the data done, we will next look to explore how to make use of machine learning models to better predict results. This report will discuss more on choosing the models and deciding which one performs better to give more reliable results.

3. Modelling Approach

3.1 Models Proposed

Based on our understanding of the AMEX dataset and the insights gathered from the EDA, the team considered several modelling approaches, including linear models, neural networks, and tree-based ensembles.

We decided to exclude neural networks for this project because the highly sparse, anonymised, tabular nature of the AMEX dataset is well-known to favour gradient-boosted decision trees over MLPs. Neural networks would require extra data processing to handle missing entries, as well as taking longer to train. We also omit logistic regression because the

data relationships are strongly non-linear, as confirmed by our EDA. Thus, tree ensembles are the most appropriate modelling family.

Among these, Random Forest and LightGBM were identified as strong candidates due to their robustness, scalability, and proven performance on similar datasets.

3.2 Models Used

After evaluating the options, the team decided to proceed with two models for the remainder of the project:

Random Forest and LightGBM

These models provide a balanced comparison between a classic ensemble baseline (Random Forest) and a more advanced gradient-boosted tree approach (LightGBM).

4. Data Preprocessing

4.1 Analysis

One of the biggest challenges to solve for the dataset was ensuring proper data preparation is performed. The initial training data provided was approximately 17GB, a filesize large enough to potentially cause memory issues if run without any processing.

We will work towards reducing the space needed to read the file. With further analysis, it can be seen that we can reduce the data size through reducing the space of each column as the dataset does not require so much space. Certain features can be compressed by changing their data type:

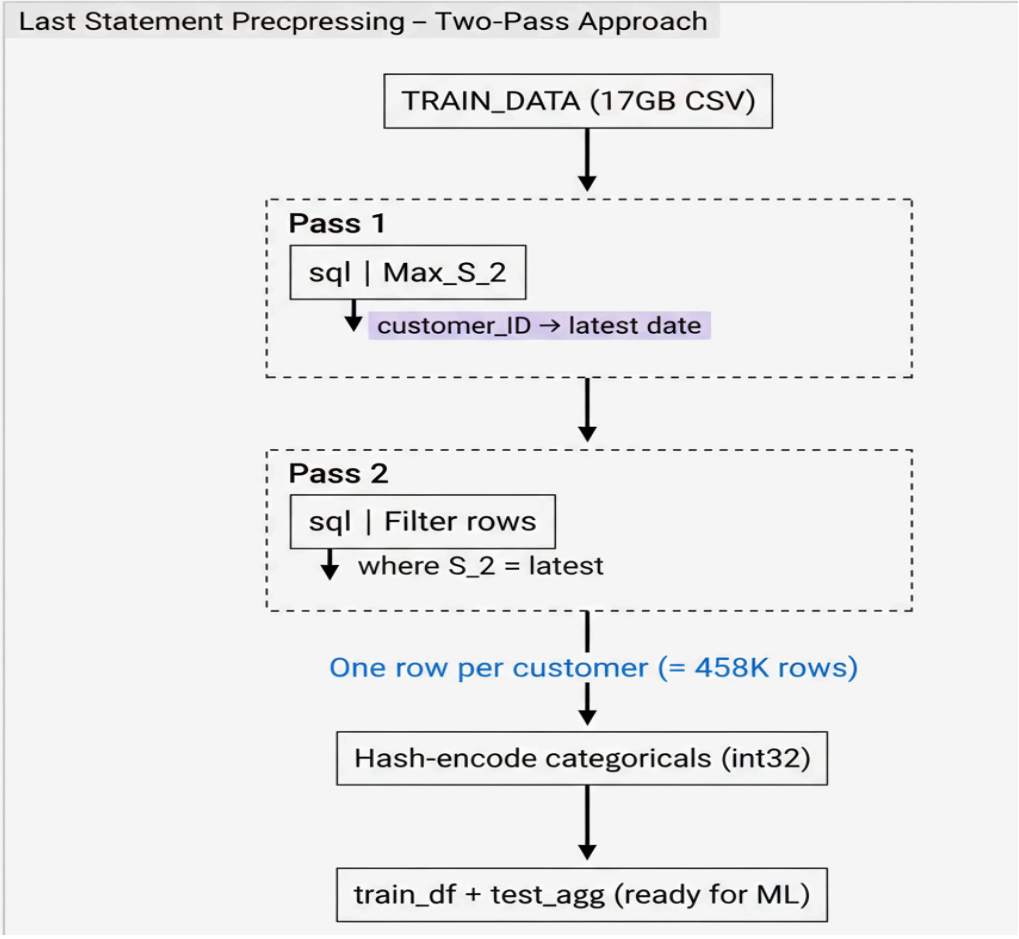
Data Column	Original Size	Reduced Size
S_2	10 bytes	3 bytes (Datetime object)
B_30	8 bytes (int64)	1 bytes (int8)
B_38	8 bytes (int64)	1 bytes (int8)
D_114	8 bytes (int64)	1 bytes (int8)
D_116	8 bytes (int64)	1 bytes (int8)
D_117	8 bytes (int64)	1 bytes (int8)
D_120	8 bytes (int64)	1 bytes (int8)
D_126	8 bytes (int64)	1 bytes (int8)
D_63	8 bytes (int64)	1 bytes (int8)
D_64	8 bytes (int64)	1 bytes (int8)
D_66	8 bytes (int64)	1 bytes (int8)
D_68	8 bytes (int64)	1 bytes (int8)
B_31	8 bytes (int64)	1 bytes (int8)

Through this, we can observe that the number of bytes used to store the data has been cut significantly.

Moreover, Parquet has also been used to cache the data. It is a filetype designed to help ensure a more efficient storage and processing of large datasets, especially for analytical queries. In this case, it is extremely useful to compress the millions of entries in the AMEX dataset.

4.2 Our Unique Solution

We work with a very large longitudinal dataset (≈17GB, multiple monthly statements per customer), so a naïve full aggregation would easily exceed memory limits. To make the problem tractable, we use a **two-pass "last-statement" pipeline:** first, we scan only `customer_ID` and `S_2` to find the latest statement date for each customer; then we scan the file again and keep only the rows whose `S_2` matches that latest date, giving exactly one "final state" row per customer. Categorical columns such as `D_63` and `D_64` are hash-encoded into compact integer codes instead of one-hot vectors, which keeps the feature matrix small and works consistently for both train and test. The result is a pair of clean tables, `train_df` and `test_agg`, that fit comfortably in memory and can be used directly by our Random Forest and LightGBM models.



4.2 Data Splitting

After constructing the final per-customer feature table (`train_df`) and merging it with the target labels, we split the dataset into a **training set (80%)** and a **validation set (20%)** using scikit-learn's `train_test_split`. We used a fixed random seed to ensure that both Random Forest and LightGBM were evaluated on the exact same validation data. This keeps

the comparison fair and allows us to understand performance differences based solely on the models themselves rather than on different data partitions.

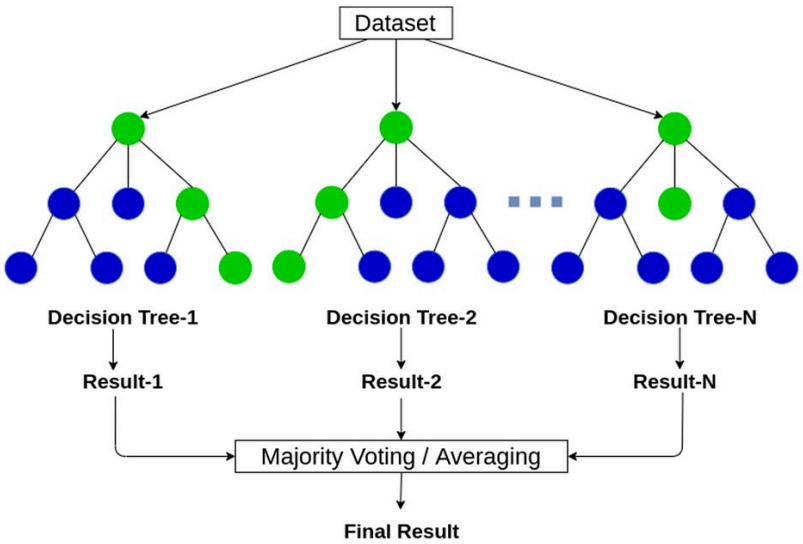
We did not use k-fold cross-validation because of the large dataset size and the computational cost involved. However, the 80/20 split provides a sufficiently large validation set to give a reliable estimate of each model's performance under the AMEX metric. Both models—Random Forest and LightGBM—were trained on the **same training features** and **tested on the same validation features**, ensuring a consistent and unbiased evaluation framework.

5. Model Training and Evaluation

5.1 Model 1 (Random Forest)

We initially attempted to use a Random Forest model, as covered in the lectures. It is a non-parametric model, meaning it will be able to capture non-linear relationships in the data. Generation of multiple decision trees makes it more resistant to overfitting and noise compared to single decision trees.

The figure below shows how a random forest is able to make decisions. For instance, it splits the ***customer_ID*** using the different column data based on their respective values. This ensures that a well and unbiased spread can be achieved. After that, it takes the average of each tree's value to ultimately decide on the final value, which is to determine the probability that the user of the AMEX card will default.



Overall, for a task such as this, with a large-scale dataset, Random Forest should serve as an effective, stable benchmark to compare future models against.

Model Configuration

Typical hyperparameters:

```
RF_PARAMS = dict(
    n_estimators=400,
    max_depth=28,
    max_features="sqrt",
    min_samples_split=4,
    min_samples_leaf=1,
    bootstrap=True,
```

```
    n_jobs=-1,
    class_weight="balanced",
    random_state=RANDOM_STATE,
)
```

We attempted both sklearn's CPU-based RandomForest and RAPIDS cuML's GPU-accelerated RandomForest. The cuML version significantly reduced training time but did not improve accuracy, likely due to the limited depth of trees needed for such a high-dimensional dataset.

5.2 Model 2 (LightGBM)

LightGBM natively handles missing values, supports categorical splitting, and scales extremely well with 180+ numerical features — all of which align with the AMEX dataset's structure.

Model Configuration

Typical hyperparameters (based on the notebook):

```
params = {
    "objective": "binary",
    "metric": "binary_logloss",
    "boosting_type": "gbdt",
    "learning_rate": 0.05,
    "num_leaves": 64,
    "max_depth": -1,
    "feature_fraction": 0.8,
    "bagging_fraction": 0.8,
    "bagging_freq": 5,
    "lambda_l1": 2,
    "lambda_l2": 1,
    "min_data_in_leaf": 50,
    "device": "gpu"
}
```

Training Process includes :

- Used **early stopping** (100 rounds) to prevent overfitting.
- Model trained on GPU using **LightGBM.Dataset()** objects.
- Custom AMEX evaluation metric implemented as a callback for validation.

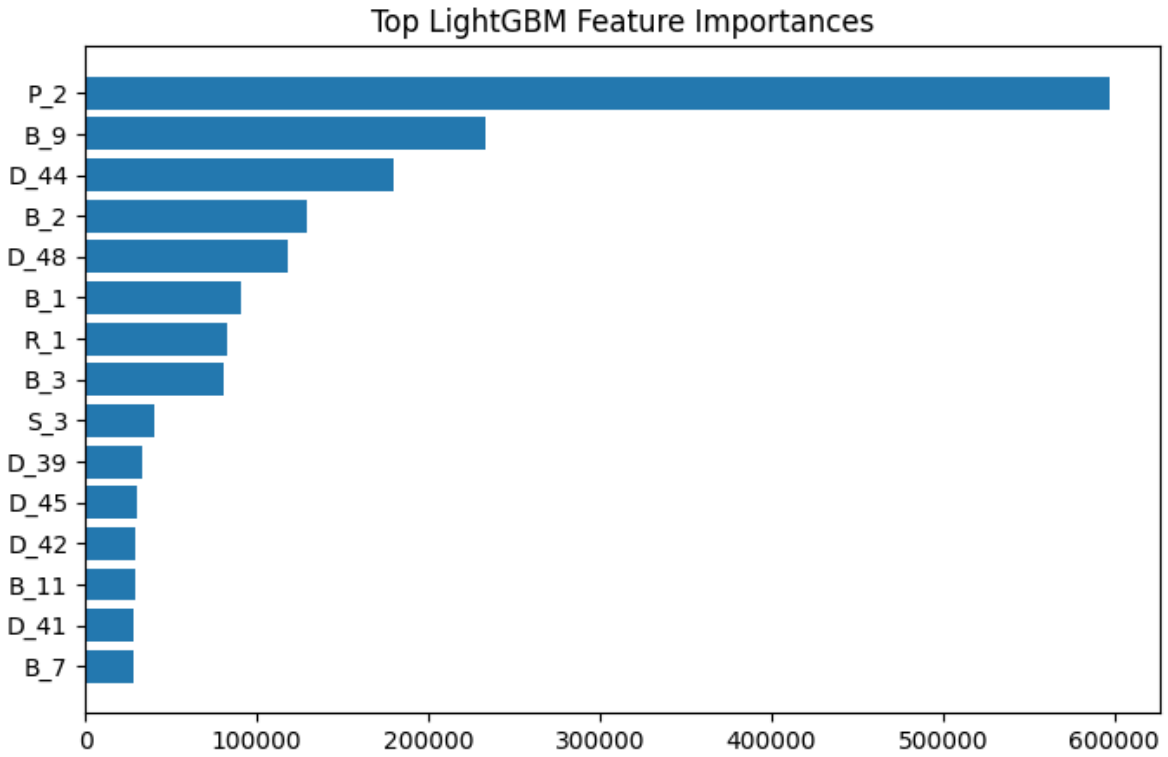
Model Evaluation

- Evaluated using:

    - **AUC (ROC-AUC)** and **custom AMEX metric** for leaderboard compatibility.

- Feature importance plotted to interpret key predictors (e.g., B_11_mean, D_39_last, R_1_std).

- Model performance compared against baseline **Random Forest** (CPU and cuML versions).



6. Results & Comparison

After running both models, we obtain the following results from the Kaggle competition.

Model	Validation AMEX	Kaggle Competition Public Score	Notes
Random Forest	0.76	0.75216	Baseline, Non-Linear but shallow.
LightGBM	0.79	0.78484	Best performing, captures nonlinear interactions

We are able to observe that the LightGBM model provides a superior performance compared to Random Forest. Listed below are the many factors that may play a role in its advantage.

6.1 Dataset Characteristic

It can be observed that this AMEX dataset is much more longitudinal in size, high sparsity with a huge imbalance in the population of data.

Below are some characteristics observed from the AMEX dataset.

a. Many time-varying features
b. Strong temporal decay features
c. Crucial Last month features

Such a huge dataset proved to be challenging for a Random Forest model. It was also less able to efficiently capture time directionality data as compared to a LightGBM model because Random Forest excels more when it comes to spatial data.

Moreover, it can be seen that Random Forest models struggle with the higher imbalance occurring in the dataset. LightGBM on the other hand is able to better mimic leaf node growth of dataset much better, allowing it to make more mature predictions.

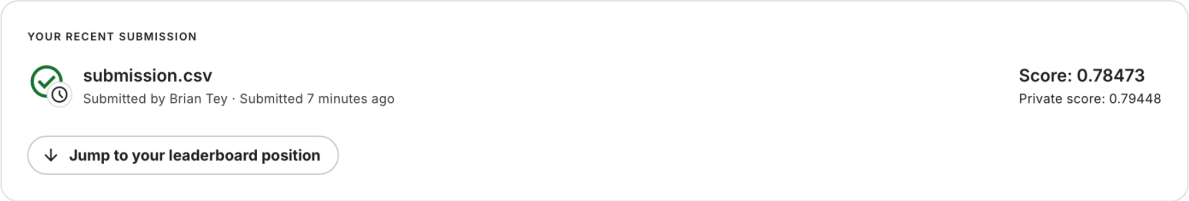
6.2 Linear vs Non-Linear

Random Forest also generally deals better with linear data as it can only compute weaker data split patterns and feature interactions are limited.

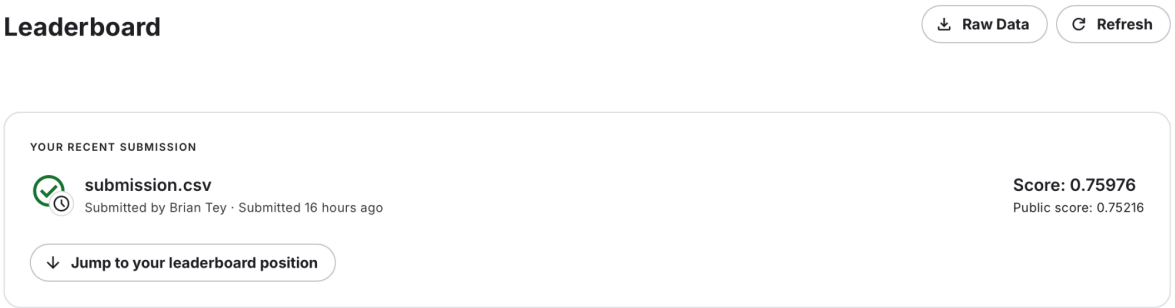
LightGBM benefits from leaf-wise growth, which discovers long chains of interactions across features such as B_11, D_39 and R_1 that Random Forest can only learn shallowly because its splits are randomized and depth-limited. This allows LightGBM to model rare but important behaviours related to default risk.

Moreover, the last-statement representation is inherently sequentially biased — the most recent financial behaviour matters most. LightGBM's boosting mechanism captures this subtle signal far better than the averaged structure of Random Forest.

6.3 LightGBM Results



6.4 Random Forest Results



6.5 Leaderboard Results

American Express - Default Prediction Late Submission ···

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

3730	kawaguc	0.78489	3	3y
3731	zukachan	0.78488	2	3y
3732	Fintech-Team	0.78487	2	3y
3733	itexname	0.78479	1	3y
3734	kleinholz	0.78476	2	3y
3735	mt05160516	0.78476	4	3y
3736	suc123	0.78473	1	3y

Our public score for LightGBM was 0.78473 but it is not showing in the leaderboard. So according to the previous rankings our ranking for LightGBM public score should be updated as 3735.

Competition Notebook	Private Score	Best Score
American Express - Default Prediction	0.79448	0.79448 V26

Search

American Express - Default Prediction Late Submission ···

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

3731	zukachan	0.78488	2	3y
3732	Fintech-Team	0.78487	2	3y
3733	itexname	0.78479	1	3y
3734	kleinholz	0.78476	2	3y
3735	mt05160516	0.78476	4	3y
3736	suc123	0.78473	1	3y
3737	fcoa	0.78473	3	3y
3738	__ka1to0607__	0.78473	1	3y
3739	Amit Maurya	0.78473	4	3y

7. Conclusion

In conclusion, this report takes a comprehensive look at the difference in performance between both Random Forest and LightGBM as well as the factors that affect their performance. Through these results, we can see that LightGBM is able to perform much better due to the nature of the dataset as well as the ability for LightGBM to capture the complex nature of the data with greater accuracy.

8. Reflection

Over the course of this project, we faced several practical challenges, such as working with a very large dataset (requiring the use of new tools like Parquet), dealing with missing information, and making sure our code could run within Kaggle's limits. As a team, we had to divide the work, discuss different ideas, and agree on a final approach, which helped us improve our communication and coordination.

Through this process, we learned that how we prepare the data and choose our models can have a big impact on the final results. Comparing Random Forest and LightGBM showed us that even when two models use the same data, they can behave very differently. Overall, the project gave us hands-on experience with the full machine-learning workflow and taught us how to work together to turn a messy, real-world problem into a working solution.

9. Future Improvements

- Incorporate time-series features (trends, deltas, rolling stats)
- Try CatBoost (handles categorical features exceptionally well)
- Experiment with feature selection or SHAP-based pruning
- Train a larger LightGBM model with monotonic constraints
- Attempt sequence models (GRU, TabNet) after reducing dimensions

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Learning from Imbalanced Data

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